



150 mph without a driver

Indy autonomous cars gear up for race, first time ever that autonomous vehicles will be raced on a circuit. <https://digit.in/nov21-25>



Hertz orders 100,000 Teslas

Hertz placed an order for 100,000 Teslas in the first step of a plan to electrify its rental-car fleet. <https://digit.in/nov21-26>

necting nerve cells, give the brain enough information to interpret and name colours.

HOW DO OUR COMPUTERS PERCEIVE COLOUR?

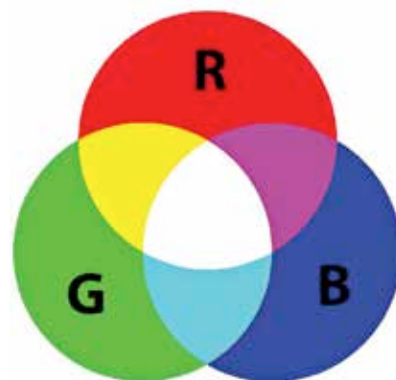
Okay, enough of the science stuff. Now let's move on to the actual fun part. You see, in the olden times we didn't have colour screens. Every display could only display black and white colour and some combinations of them too. But none of the displays could display any colour except black or white. So, how do we actually go around displaying colour on our displays? Well, just like everything in a computer, a colour image too is a collection of Os and 1s. Those Os and 1s are arranged in a very specific way. The use of RGB to define an image's colours has become pretty standard by now but what is RGB and why is it so common? Well, RGB simply stands for Red – Green – Blue. Scientists and computer engineers figured out that almost every colour that the human eye can see can be represented in a specific combination of Red, Green and Blue values. Hence, if we add just the right amount of Red, Green and Blue we can get whatever colour we want at our disposal and that is exactly why image data or the colour value of a particular pixel is stored in the form of RGB. There's a very important concept of bit-depth here but that'll be covered later in the article.

Now that we know how to display a particular colour on a screen, it's time for us to deploy it. We can apply this RGB formula for every camera and every display on Earth and everything will be good and dandy, right? We won't need to do anything apart from this, right? Well, we wish it were that simple.

Have you ever gone to buy a TV or a monitor in a shop and you've seen many displays displaying the same thing? Most of the time it is some promotional material about their company but have you noticed that even though the same video is being

played on all the displays, the image still looks a bit different on all of them. It's like every TV is displaying a different version of that image. In order to minimise this effect, we need colour spaces and most importantly, colour gamuts.

Just because every camera lens captures an image differently, every person perceives colour differently and every display can display colour differently, we need a standard that everyone can follow



FACTS ABOUT COMMONLY USED COLOR SPACES!

sRGB

- sRGB is the most common colour space in the world. The whole web uses it and almost all phones and displays use it too.
- Making your content in sRGB is a sure shot way of knowing that your content will look great on the web.
- It was published on 5th November 1996 and is being heavily used till date.

Adobe RGB

- While the web uses sRGB, Adobe RGB has become a standard for printing stuff.
- Almost all printers today use Adobe RGB as their default colour space just because it can cover more colours than sRGB and colours look more vibrant on it.
- It was created in 1998 by Adobe systems with the idea of encompassing most colours achievable by CMYK printers.

Rec. 709

- The Rec. 709 and the sRGB standard share almost the same colour space on the LAB gamut.
- This is where the similarities end. The Rec. 709 is a newer colour space developed to suit newer displays.
- It features a different gamma value than sRGB. Gamma basically defines how colour spaces encode and decode colour information.
- The Rec. 709 colour space is widely used in HDTVs (1080p) today.

Rec. 2020

- The Rec. 2020 is a relatively new colour space developed for UHD (4K and 8K) TVs.
- It is a much bigger colour space as compared to Adobe RGB, sRGB and even Rec. 709.
- Rec. 2020 covers 75.8% of the whole LAB gamut whereas the sRGB covers only 35.9%.

so that content can look almost the same on every display.

This is exactly why we need a colour gamut and colour spaces to assist us.

COLOUR GAMUTS AND COLOUR SPACES

Coming to the main topic of this article, colour gamuts are basically mathematical representations of a colour range onto a graph. Too complicated? Well, you can understand it like this – take every colour that a human eye can possibly see. Mathematicians took that and derived an equation for it. They plotted that equation on a graph. That is a colour gamut. Take all the colours that a particular device can reproduce and plot it on a graph and you've just got yourself a colour gamut.

If you've done some research about colours and colour grading videos, you must've come across the diagram on the next page. This is the colour